

Environment Setup (What to mention)

Option A: Linux (Recommended)

- OS: **Ubuntu 22.04 LTS**
- Platform: **VirtualBox**
- User type: **Standard user (non-root)**

Option B: Windows

- OS: Windows 10 / 11
- Security Tool: Windows Defender + Windows Firewall

 *Take screenshots if possible (terminal + firewall enabled)*

User Accounts & Privileges

Linux Commands Used

whoami

id

cat /etc/passwd

groups

Key Points to Document

- Linux supports **multi-user environment**
- Users are divided into:
 - **Root user** → full control
 - **Normal user** → limited permissions
- Root access is obtained using:

sudo su

Security Concept:

Admin privileges should be limited to reduce damage if an account is compromised.

File Permissions (IMPORTANT)

Check Permissions

```
ls -l
```

Example Output

```
-rwxr-xr--
```

Permission Breakdown

Symbol Meaning

r Read

w Write

x Execute

Entity Description

Owner File creator

Group Group users

Others Everyone else

Change Permissions

```
chmod 700 file.txt
```

```
chmod 644 file.txt
```

Change Ownership

```
sudo chown user:user file.txt
```

Security Reason:

Prevents unauthorized file access or execution.

Firewall Configuration

Linux – Enable UFW

```
sudo ufw status
```

```
sudo ufw enable
```

```
sudo ufw allow ssh
```

```
sudo ufw deny 23
```

Verify

```
sudo ufw status verbose
```

Windows – Firewall

- Open **Windows Security**
- Enable:
 - Domain Firewall
 - Private Firewall
 - Public Firewall

Why Firewall?

- Blocks unauthorized inbound/outbound traffic
 - First line of defense
-

5 Running Processes & Services

View Running Processes

```
ps aux
```

```
top
```

```
htop
```

List Services

```
systemctl list-units --type=service
```

Check Specific Service

```
systemctl status ssh
```

6 Disable Unnecessary Services

Example

```
sudo systemctl stop bluetooth
```

```
sudo systemctl disable bluetooth
```

Security Benefit

- Reduces attack surface
 - Fewer services = fewer vulnerabilities
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7 OS Hardening Best Practices (CORE DELIVERABLE)

OS Security Checklist

☒ Use strong passwords

☒ Disable root login

☒ Enable firewall

☒ Apply system updates regularly

```
sudo apt update && sudo apt upgrade
```

☒ Remove unused packages

```
sudo apt autoremove
```

☒ Least privilege principle

☒ Disable unnecessary services

☒ Monitor logs

```
journalctl
```

☒ Enable automatic updates

☒ Use antivirus (Windows Defender / ClamAV)